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1) Difference between coupling and cohesion.

The major difference between cohesion and coupling is that cohesion deals with the interconnection between the elements of the same module. But, coupling deals with the interdependence between software modules.

2) Enlist two points on software design

1 Overview · 2 Design concepts · 3 Design considerations · 4 Modeling language ·
5 Design patterns · 6 Technique · 7 Usage

3) Enlist four risk software development phase

For most software development projects, we can define five main risk impact areas:

i) New, unproven technologies. (ii) User and functional requirements. (iii) Application and system architecture. (iv) Performance. (v) Organizational.

4) Write three parameters of collaboration diagram?

(i) Objects (ii) Links (iii) Messages

5) Thumb rule

Who I Know - Rules of Thumb

☐ an actor knows about its participants

Person knows about cashier

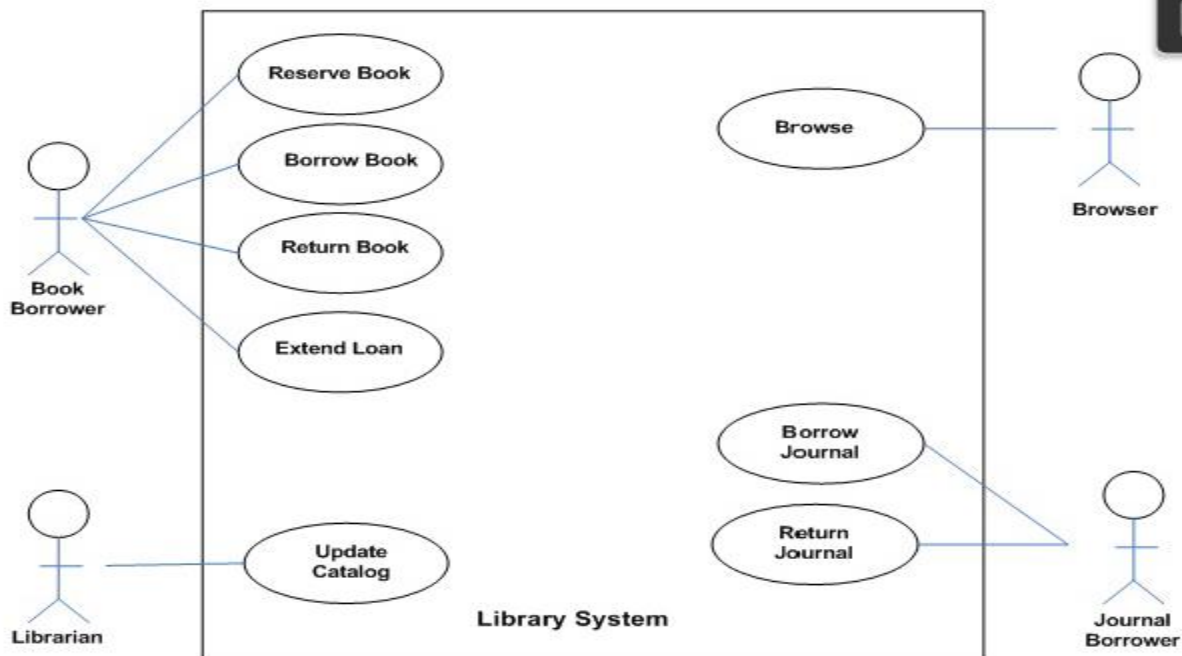
☐ a transaction knows about its participants

A session knows about its register and cashier

☐ A transaction contains its transaction line items

Sale contains its sales line items

6) Library diagram



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7) Identify Class, Objects and Relationships

Alia has a fruit basket

Objects: Alia, Fruit, Basket

Classes: Person, Fruits, Baskets

Relations: Alia ---- Fruits ----- Association

Alia ---- Basket ----- Association

Basket -- Fruits ----- Association – Container

8) System engineering are assigned the task of developing system models which process must be included in system methodology

Systems Engineers are responsible for assessing systems, determining problems, providing solutions to issues that arise, designing systems, upgrading systems, maintaining systems and brainstorming possible improvements that can be made to a system in the future.

9) State, behavior of an object

State of an object - The state or attributes are the built in characteristics or properties of an object. For example, a T.V has the size, colour, model etc.

Behaviour of the object - The behavior or operations of an object are its predefined functions. For example, a T.V.

10) Types of functional requirements

Some of the more typical functional requirements include:

- Business Rules.
- Transaction corrections, adjustments and cancellations.
- Administrative functions.
- Authentication.
- Authorization levels.
- Audit Tracking.
- External Interfaces.
- Certification Requirements.

11) How collaboration diagram better than sequence diagram

Sequence diagrams are best to see the flow of time. On the other hand, static object connections are best represented by collaboration diagrams. Sequence of messages is more difficult to understand in collaboration diagrams than in the case of sequence diagrams. On the other hand, object organization with control flow is best seen through collaboration diagrams. It may be noted that complex control is difficult to express anyway but collaboration diagrams can become very complex very quickly

12) What are the post-condition? Student login and the schedule of lectures

1. Where to look. 2. What to look for 3. How to design and implement program 4. How to test program.

13) Software phases Enlist the names of phases

- Planning Stage

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- Feasibility or Requirements of Analysis Stage
- Design and Prototyping Stage
- Software Development Stage
- Software Testing Stage
- Implementation and Integration
- Operations and Maintenance Stage

14) Functional Requirements:

A Functional Requirement (FR) is a description of the service that the software must offer. It describes a software system or its component. ... It can be a calculation, data manipulation, business process, user interaction, or any other specific functionality which defines what function a system is likely to perform

15) Non-Functional Requirements:

In systems engineering and requirements engineering, a non-functional requirement (NFR) is a requirement that specifies criteria that can be used to judge the operation of a system, rather than specific behaviors. They are contrasted with functional requirements that define specific behavior or functions.

16) Aik or scenario tha us mai actors btany thy?

Select actors

Actors are people and organizations that take part in the system under consideration.

Examples of actors are: person, organization (agency, company, corporation, foundation).

17) Object-Oriented Analysis using Abbot's Textual Analysis

Part of speech = proper noun, improper noun, doing verb, being verb, having verb, adjective, adjective phrase

Model component = instance, class/type/role, operation, classification, composition, attribute value or class, association, operation

Example = Mehdi Hassan, (student, teacher), buy, (is a horse, is a book), fan has wings, this ball is green, the customer with children, the customer who bought the kite

18) Technique for requirement data management

The data requirements analysis process employs a top-down approach that emphasizes business-driven needs, so the analysis is conducted to ensure the identified requirements are relevant and feasible. The data requirements analysis process consists of these phases:

1. Identifying the business contexts
2. Conducting stakeholder interviews
3. Synthesizing expectations and requirements
4. Developing source-to-target mappings

19) In object-oriented programming, collaboration occurs among objects that mimic real-world objects inside an application.

20) Three Types of Collaboration

- Creative – where two or more people create something to achieve a specific goal,

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- Connective – bringing together information from disparate sources, and.
- Compounding – where a team builds on previous achievements to reach further.

21) Software architecture.

The design process for identifying the sub-systems making up a system and the Framework for sub-system control and communication is architectural design. The output Of this design process is a description of the software architecture.

22) Software development

The activities involved in software development can broadly be divided into two major categories first is construction and second is management. The construction activities are those that are directly related to the construction or development of the software.

23) The Balancing Act!

These requirements may be conflicting. For example, there may be tension among the Following:

- ☐ Cost vs. Efficiency
- ☐ Cost vs. Reliability
- ☐ Efficiency vs. User-interface

A Software engineer is required to analyze these conflicting entities and tries to strike a Balance

24) Define whole part aggregation and inheritance

Aggregation. A special form of association that specifies a whole-part relationship between the aggregate (whole) and a component part. Composition. A form of aggregation with strong ownership and coincident lifetime as part of the whole.

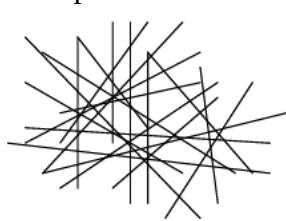
Inheritance, also called succession, the devolution of property on an heir or heirs upon the death of the owner. The term inheritance also designates the property itself

25) Attributes of cinema ticketing system

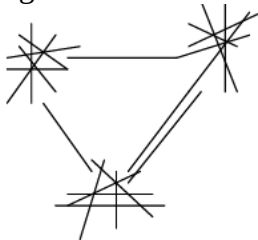
Movie Ticket Booking System entities and their attributes :

Booking Entity : Attributes of Booking are booking_id, booking_title, booking_type, booking_ticket, booking_date, booking_description. Payment Entity : Attributes of Payment are payment_id, payment_customer_id, payment_date, payment_amount, payment_description.

26) example of low and high coupling

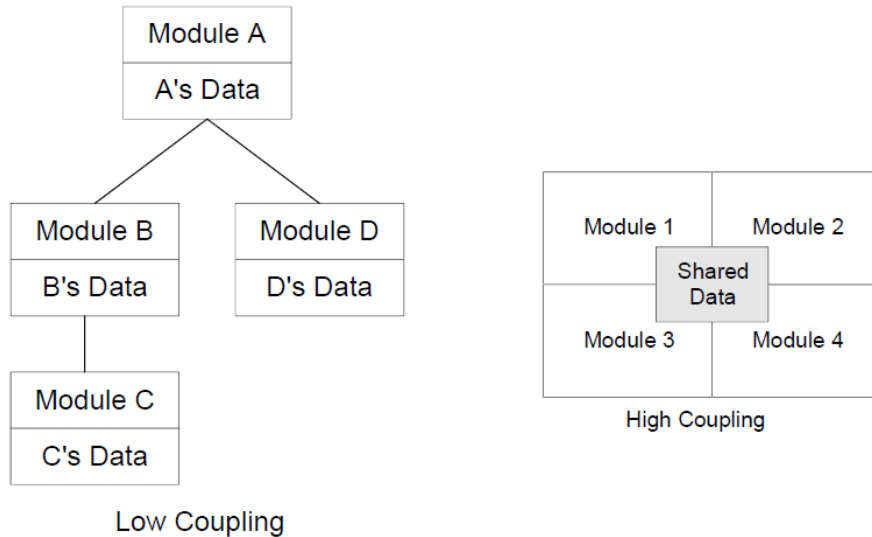


High Coupling

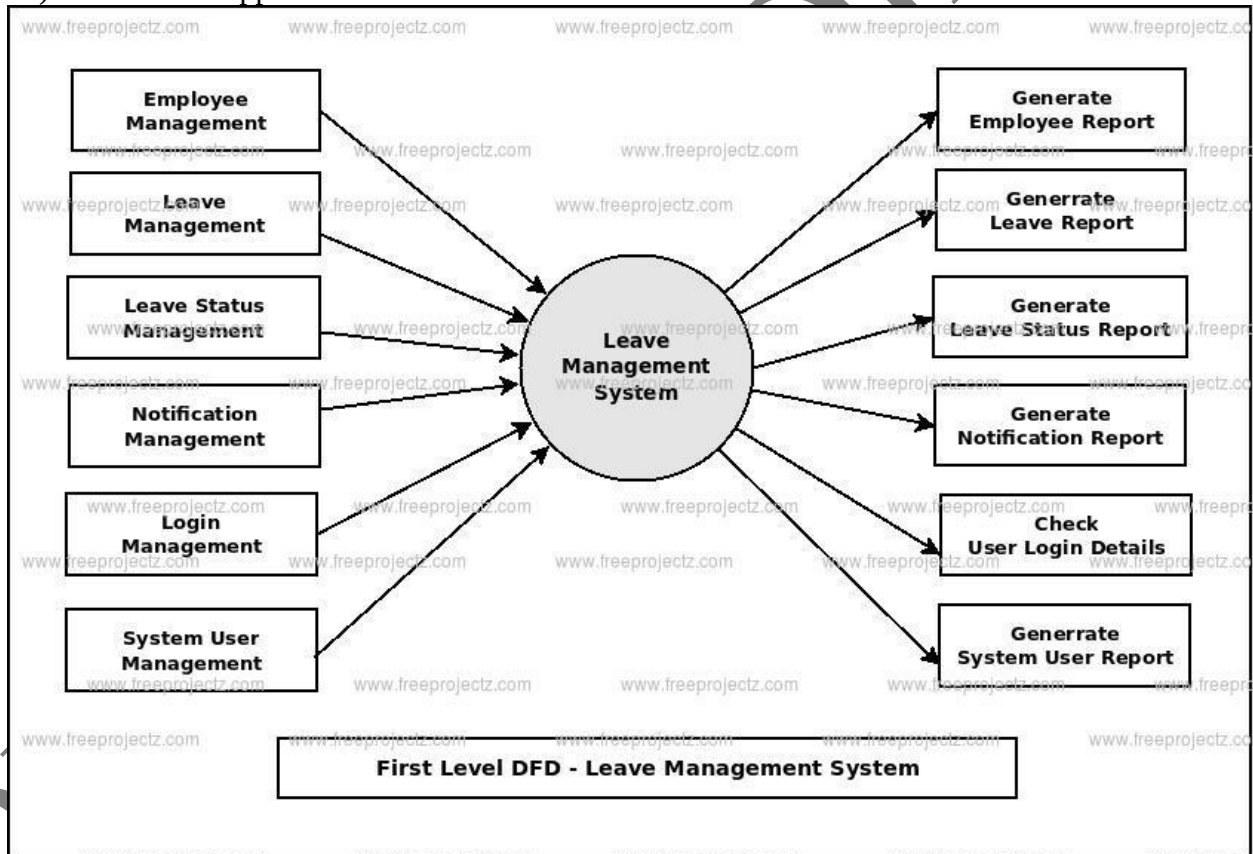


Low Coupling

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27) Dfd of leave application



28) Write two fundamental reason why software architecture is important

1. Defining a solution to meet requirements
2. Enabling and inhibiting quality attributes

29) What are the main parts of software?

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Your system has three basic types of software: application programs, device drivers, and operating systems. Each type of software performs a completely different job, but all three work closely together to perform useful work.

30) Activities of software construction phase

Construction

The construction activities are those that directly related to the development of software, E.g. Gathering the requirements of the software, develop design, implement and test the Software etc. Some of the major construction activities are listed below.

☐ Requirement Gathering ☐ Design Development ☐ Coding ☐ Testing

31) Activities of design phase

In the design phase, one or more designs are developed, with which the project result can apparently be achieved. Depending on the subject of the project, the products of the design phase can include dioramas, sketches, flow charts, site trees, HTML screen designs, prototypes, photo impressions and UML schemas

32) scenerio given tha us m kuch activities mention thi batana tha k ye software engineering ka konsa phase h(long)

Software Engineering Phases

There are four basic phases of software development

Vision: Here we determine why are we doing this thing and what are our business objectives that we want to achieve.

Definition: Here we actually realize or automate the vision developed in first phase. Here we determine what are the activities and things involved.

Development: Here we determine, what should be the design of the system, how will it be implemented and how to test it.

Maintenance: This is very important phase of software development. Here we control the change in system, whether that change is in the form of enhancements or defect removal.

33) Coding k ilawa aur konse cheezy hoti h jo software bnane me important hoti hn(5 marks)

Software development is the process of conceiving, specifying, designing, programming, documenting, testing, and bug fixing involved in creating and maintaining applications, frameworks, or other software components.

34) fact of peter coad methodology

The principle that any operation that achieves a well-defined effect can be treated by its users as a single entity, despite the fact that the operation may actually be achieved by some sequence of lower-level operations.

35) Reasons to choose the prototyping at least 2

There are many great reasons to make prototypes and to include them early and often in your process, and the four main points i'll cover are as follows: to understand, to communicate, to test and improve, and to advocate

36) Role of Architectural design

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Architectural Designers plan and design the layouts for buildings and landscapes. They work under the direct supervision of licensed Architects and interact with clients to gather design requirements, specifications, and budgets for their building projects.

37) Question on software engineering framework likely to be asked

Software Engineering Framework

Quality Focus: The quality focus demands that processes be defined for rational and timely development of software

Processes: The processes are set of key process areas (kpas) for effectively manage and deliver quality software in a cost effective manner

Methods: Methods provide the technical “how-to’s” to carryout these tasks

Tools: Tools provide automated or semi-automated support for software processes, methods, and quality control.

38) Data Flow Diagram (DFD) basics

DFD

- ☐ Processes on a data flow can operate in parallel.
- ☐ Looping and branching are typically not shown.
- ☐ Each process path may have a very different timing.

39) Coupling and Cohesion

Coupling is a measure of independence of a module or component. Loose coupling

Means that different system components have loose or less reliance upon each other.

Hence, changes in one component would have a limited affect on other components.

40) Object types

There are three kinds of object:

(i) Direct Object (e.g., I know him.) (ii) Indirect Object (e.g., Give her the prize.) (iii) Object of a Preposition (e.g., Sit with them.)

41) Requirement specification

Requirement Specification Characteristics

A good Requirements specification document should possess the following characteristics.

- ☐ Complete - No requirement or necessary information should be missing.
- ☐ Consistent – No requirement should conflict with other software or higher-level system or business requirements.

42) Case diagram steps

1. Identify the Actors (role of users) of the system.
2. For each category of users, identify all roles played by the users relevant to the system.
3. Identify what are the users required the system to be performed to achieve these goals.
4. Create use cases for every goal.
5. Structure the use cases.

43) Four case diagram strategies

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Ucds have only 4 major elements: The actors that the system you are describing interacts with, the system itself, the use cases, or services, that the system knows how to perform, and the lines that represent relationships between these elements.

44) Use case diagram (write that's information)

A use case diagram is a way to summarize details of a system and the users within that system. It is generally shown as a graphic depiction of interactions among different elements in a system. ... A use case is a methodology used in system analysis to identify, clarify, and organize system requirements.

45) OOD challenges

The major challenges for a software engineer is that he has to build software within Limited time and budget in a cost-effective way and with good quality

46) Type name of transactional function types

One of the three kinds of functions (with facilitating functions and logistical functions) performed by intermediaries in a marketing channel; transactional functions are the activities associated with buying products and reselling them, and the risks incurred in keeping the products in stock.

47) Define Class, Diagram, Sequence Diagram

(i) A class is a blueprint that defines the variables and the methods common to all objects of a certain kind.

(ii) A class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among objects.

(iii) Sequence Diagram is an interaction diagram that details how operations are carried out -- what messages are sent and when.

48) Components of Software while making a computer program

Your system has three basic types of software: application programs, device drivers, and operating systems. Each type of software performs a completely different job, but all three work closely together to perform useful work.

49) Radio and cat are objects, explain.

Radio and cat have state, behaviour and properties. They may belong to Electric device and animal classes respectively so we can say that these are objects.

50) Write four software development loop phases.

Software Development Loop

Problem Definition: In this stage we determine what is the problem against which we are going to develop software

Technical Development: In this stage we try to find the solution of the problem on technical grounds and base our actual implementation on it.

Solution Integration: If there are already developed system(s) available with which our new system has to interact then those systems should also be the part of our new system.

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Status Quo: After going through the previous three stages successfully, when we actually deployed the new system at the user site then that situation is called status quo.

51) Write four parameters of software quality design which can be used from different angles to balance the quality.

Cost, efficiency, reliability, maintainability.

52) Write mistake in contextual digram and solution.

Two operations can not labeled on single data flow arrow. We should draw single data flow arrow for single operation.

Practical Questions:

Aik class diagram the us main relationship identify karne the (lecture # 19)

Cohesion and coupling between electricity (lecture # 12)

Why we consider radio and cat as an object

An atm cash is withdrawal

Peter coads ki example di thi unki methodologies key name likhney they

Or ek diagram di thi us m external entities btani thi (page 58)

Or ek diagram di thi us m object relation ship (page 92)

Ek m condition di ti functional or non functional requirement batani ti

Ek long question m senerio dia ta cinema ticketing system us ke attributes waghaira likhne te

Ek long m do cases diye te ek m highly cohesive or highly coupling hai woh batana ta or dosare case m high coupling hai ya low ye batana ta

.DFD given thi us me mistake find krni thi Leave application wali thi

Ik example thi electric circuit ki us me Coupling or cohesion level btana tha

Electricity ka bataya hua tha or usma relation batana tha cohesion and coupling ka

Problem di hui thi usma select participant, place, Tangible things, actors ye sb 1 batana tha

8.2. SELECTING OBJECTS

8.2.1. Select Actors the actor is: person

8.2.2. Select Participants the Participants are:

For further please see attach file in mail

Diagram given thi high coupling and low coupling babtani thi

Diagram given thi and according to student model b best tha contradict about student remarks with two reason (Long)

Question tha k objects and behaviour find kre according to textual analysis

3) Diagram s btana tha use case and colaborat aur class vgyra 3 topic thy

Whole part wala 1 table tha (5 marks)

Tell the components of sugar factory

Lecture # 16 is most important and 12